

1. Introduction

A Background on Time Series Classification

1. Time Series Analysis

Today, there is an abundance of time series data. From power usage to ECG graphs, thousands of hours of time-dependent data is being collected constantly. Time series classification is one of many tools used to gather information by analyzing this data.

1. Explaining Classification

While a substantial amount of research has been done to find the most accurate classification algorithms, there also lies an issue in explaining them. If these algorithms are used to provide predictions that can have a human impact, such as medical diagnosis, it is essential that there is an explanation that can be easily understood. Our research question is to explain how a set of time series data is classified.

1. Choosing a Classification Method

The algorithm I chose to help me explain this is the Bag of SFA Symbols (BOSS) transformation. I chose this algorithm for two primary reasons. Firstly, it is one of the most accurate algorithms. Secondly, it divides the data into “words” that can be individually analyzed.

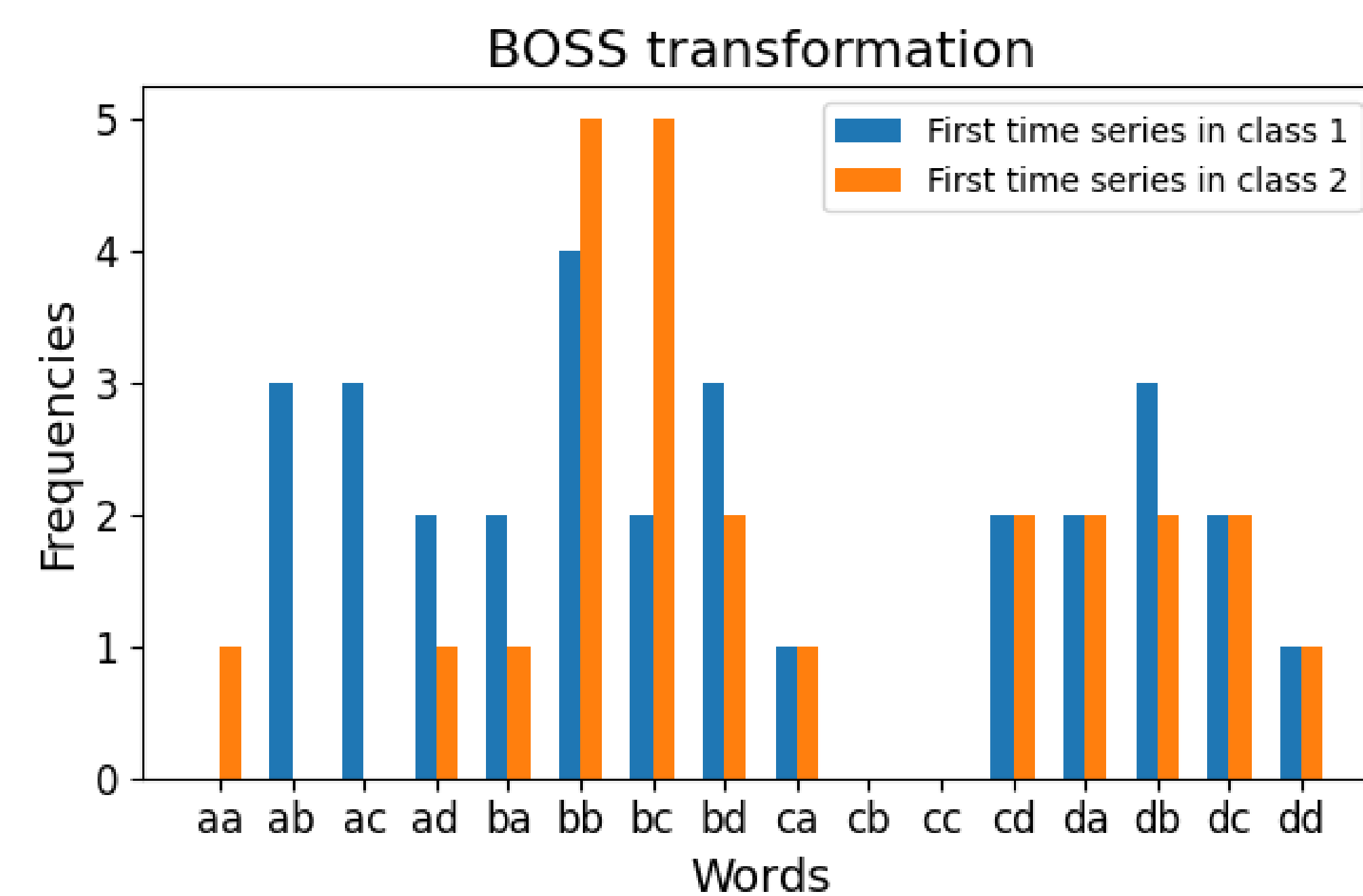
2. Transformation

How does the transformation work?

1. At the most basic level, BOSS transforms a data instance into a metaphorical bag of words, where the number of occurrences of each word is counted.

1. At its core, BOSS first breaks the data into windows. Then each window is transformed into a string of symbols through a Symbolic Fourier Approximation (SFA).

1. Finally, immediate repetitions are removed, and the number of occurrences of each “word” is counted.



3. Classification

How is the data classified?

1. The data is classified using a 1-nearest neighbour algorithm. It is the simplest variation of the k-nearest neighbours algorithm, which is very commonly used in machine learning.

1. First, the BOSS distance is calculated between the query instance and all of the reference/training instances. The query is then assigned the class of the nearest instance in the training set.

1. The BOSS distance formula is very similar to the Euclidean distance, but all values of zero in the query are ignored.

$$\text{dist}(B_1, B_2) = \sum_{a \in B_1; B_1(a) > 0} [B_1(a) - B_2(a)]^2$$

Fig. 2: Visualization of K-Nearest-Neighbour around the green point with rings for k=3 and k=5. Equation from [Schäfer, 2015]. Where B_i is the transformed BOSS data for time series T_i .

4. Results

Single Feature Knockout Tests

1. Feature Knockouts

Knockout tests are used in biology to determine the role of individual genes. Instead of knocking out a biological gene, I knocked out a feature and recorded the impact on classification accuracy.

1. Accuracy Change

The knockouts made the classification less accurate on average, but results ranged from increasing by 6.0% to decreasing by 32.0%. For individual classes, however, the results were much more varied. The accuracy changed anywhere from -52.6% to +9.5%.

1. Computation Time

Each knockout test I ran took ~1.1 min to compute. (25-30 knockouts on 150 test instances of GunPoint).

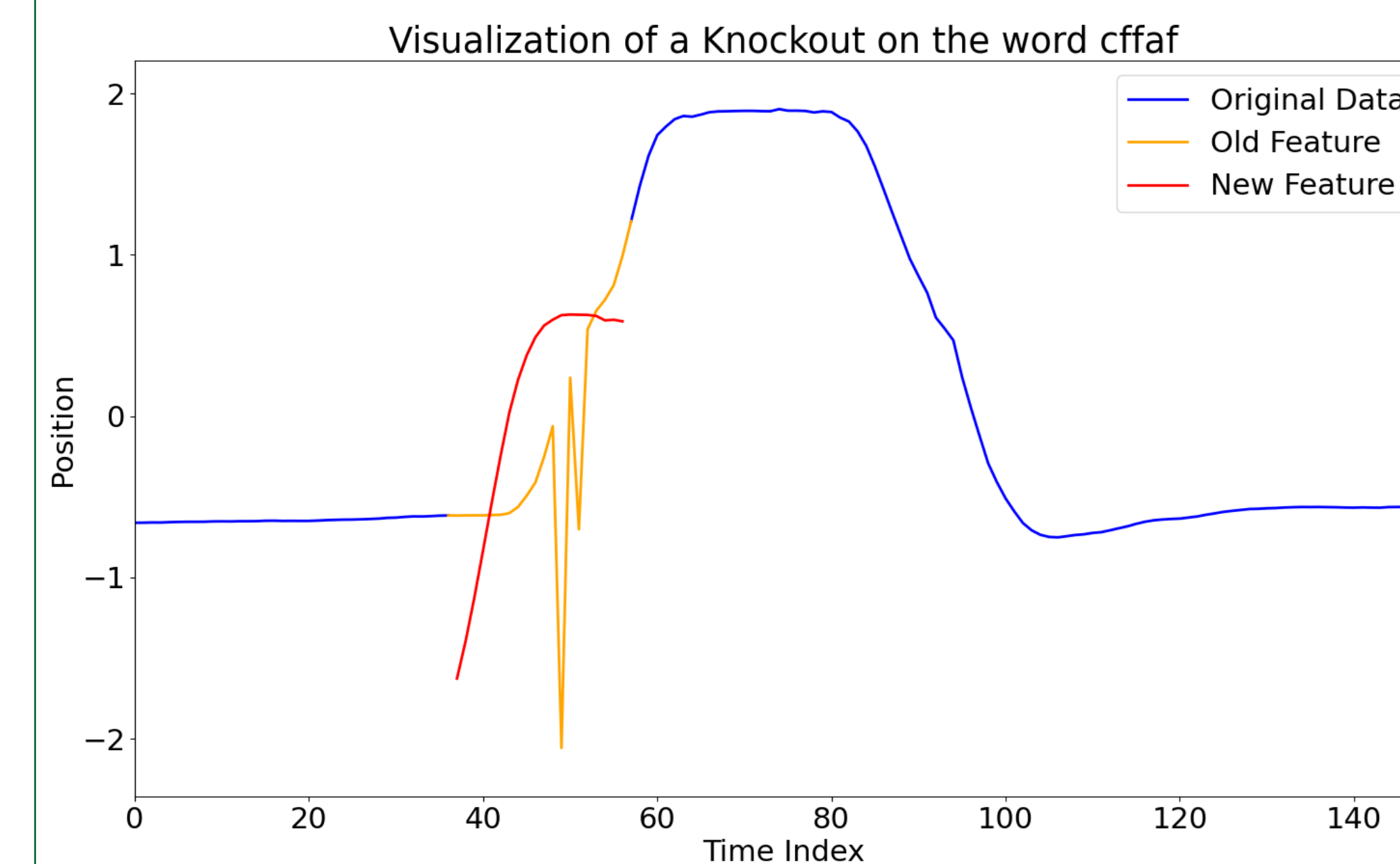


Fig. 3: A knockout of the word CFFAF from the 19th instance of the GunPoint testing data set. The replacement feature is DAFAF from the 3rd test instance. Parameters: (5, 6, 20)*.

Information Gain Calculations

1. I used information gain (IG) to determine the parts of the time series that had the most impact on classification.

1. Information Gain is a formula used in machine learning and decision trees based on entropy to determine the value of a decision node (or feature in my case).

1. A feature with an IG value of 1.0 would divide the data set perfectly into its classes. With a computation time of ~5 seconds, I was able to find features on the GunPoint data set with IG values of up to ~0.3.

5. Concluding Remarks

What can be inferred from these results?

1. Feature Analysis

There are viable methods to determine which features of a time series are most important to classification. By analyzing these individual features, it can be determined why a classification decision has been made.

1. Visuals

Through the aid of simple visuals, these explanations can be more easily understood.

1. Use Cases

Possible uses for this research would be to find errors in training data or explain medical diagnoses to patients. Another topic to explore would be how individual feature analysis can be used for data compression.

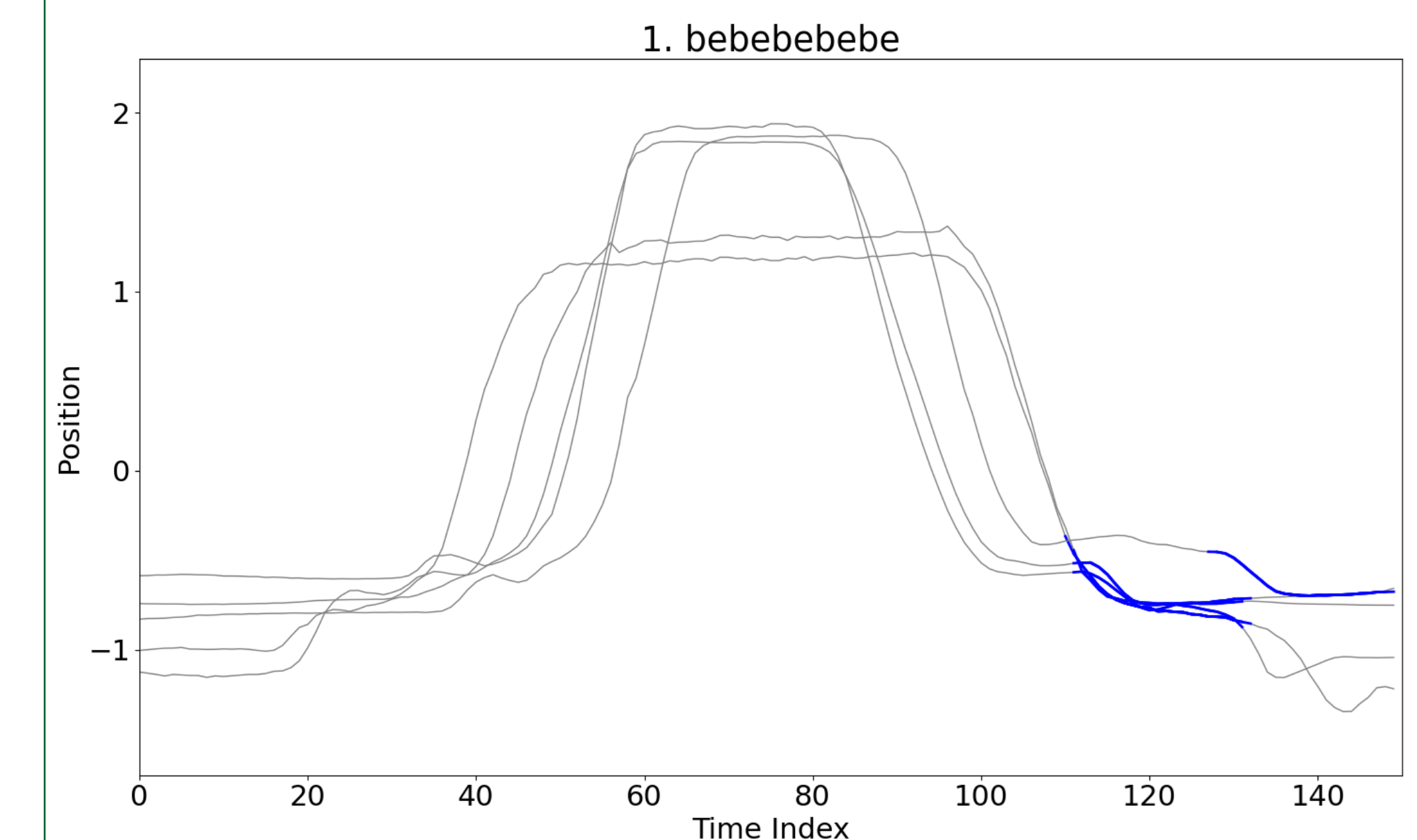


Fig. 4: The most discriminatory feature on the GunPoint data set according to IG with the parameters (10, 6, 20)*. The word is BEBEBEBEBE and its IG value is 0.3110900603007193.

6. Acknowledgements

Thank You!

1. I would like to extend my gratitude to the Tate family for sponsoring the HIP Program.

1. I would also like to thank the mentors and professors for committing their time to this program.

1. All sources for research and visuals, as well as some additional links, are listed at the bottom of the poster.